

Real-Time Light Dosimetry for Heritage Digital Twins

via Texture Space Ray Tracing

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Context and Motivation

- **Context:** Museum artifacts suffer irreversible photometric damage from cumulative light exposure over time.
- **Motivation:** Physical lux meters are spatially limited and cannot predict future damage under changing environmental conditions.
- **Need:** A highly scalable Digital Twin capable of simulating an entire 8,760-hour yearly cycle.

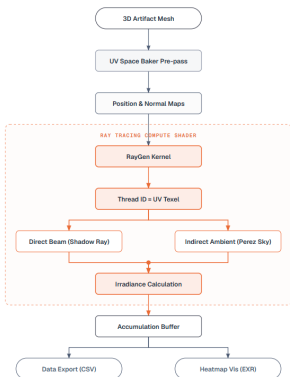
Objectives

- **Main Objective:** To study, design, and implement a real-time metrology digital twin using Texture Space Ray Tracing.
- **Secondary Objectives:**
 - Integrate astronomical equations for solar ephemeris.
 - Implement the Perez All-Weather Sky Model for indirect light.
 - Evaluate the impact of architectural conservation scenarios.

Related Work

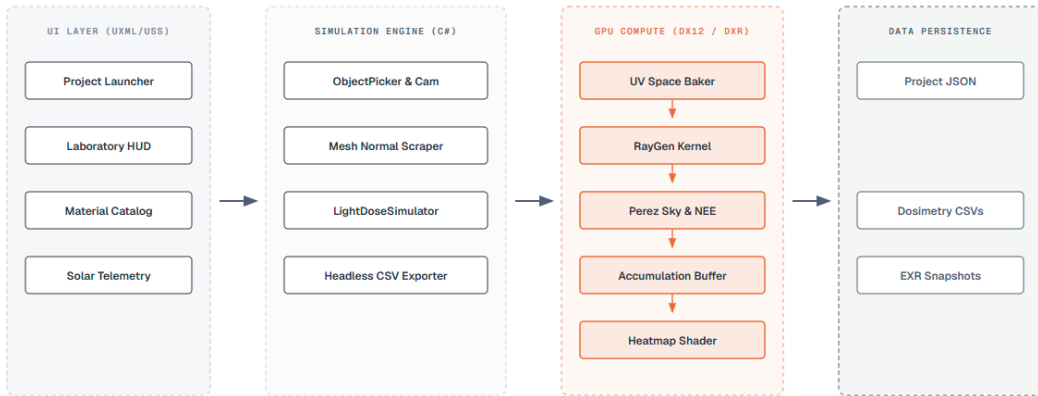
Reference	Description	Advantage/Disadvantage
Pharr et al. (PBRT)	Physically Based Rendering	Foundation of light transport mathematics
Veach & Guibas	Monte Carlo Integration	Highly accurate, but computationally expensive
Perez et al.	Sky Luminance Distribution	Predicts realistic cloudy/clear sky ratios
Traditional Screen-Space	Rasterization Techniques	Fast, but fails with off-screen geometric occlusion

Solution Overview: System Architecture



- **The Core Pipeline:** C# orchestrates the celestial telemetry on the CPU, while the hardware-accelerated DirectX 12 GPU executes the path tracing.

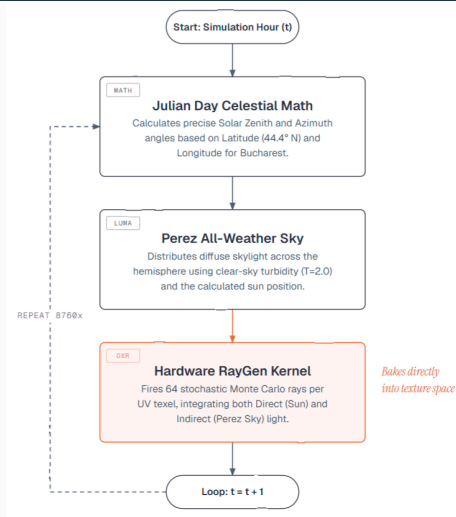
Solution Overview: Detailed Architecture



Solution: Astrometric Orchestration

- **C# CPU Thread:** Decoupled from the rendering engine to prevent viewport freezing.
- **Julian Day Math:** Converts the 8,760-hour cycle into exact geospatial sun altitudes and azimuths.
- **Result:** A completely deterministic, geographically synchronized solar vector for the GPU.

Solution: Texture Space Path Tracer

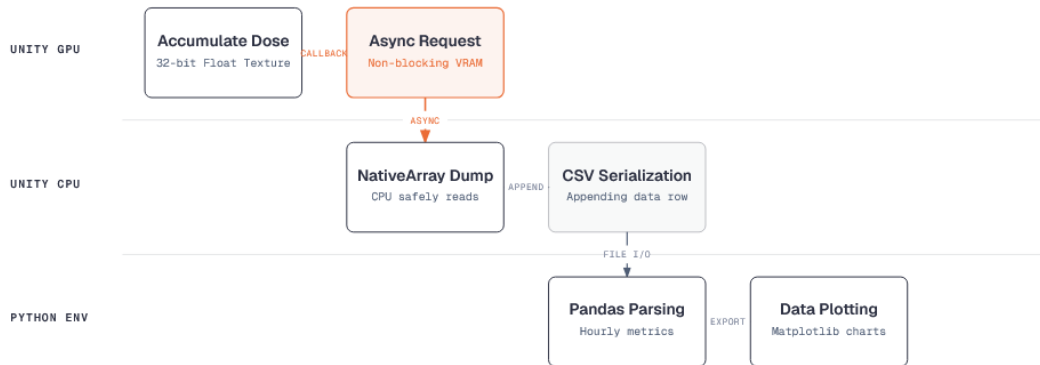


- **Bypassing Screen Space:** By baking directly into the artifact's UV maps, we ensure perfect geometric occlusion even when the camera looks away.

Solution: Ambient Skylight

- **Perez All-Weather Model:** Simulates realistic relative luminance distribution on the sky dome (Clear Sky, $T = 2.0$).
- **Cosine-Weighted Importance Sampling:** Stochastic multi-sampling concentrates rays toward the zenith, dramatically reducing variance.

Solution: Data Telemetry



- **Asynchronous Readback:** Ensures that extracting the 32-bit float accumulation

Experiments and Validation: Setup

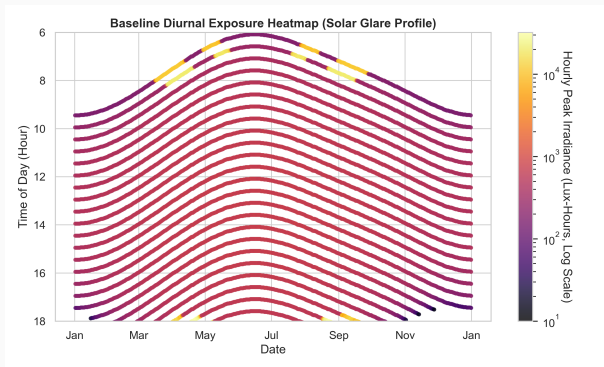
- **Asset:** The Stanford Bunny, geometrically normalized and centered.
- **Location:** Bucharest, Romania.
- **Hardware:** NVIDIA RTX 4070 Ti Super (DX12 DXR).
- **Simulation Duration:** One full meteorological year (8,760 hours).



Experiments: Conservation Scenarios

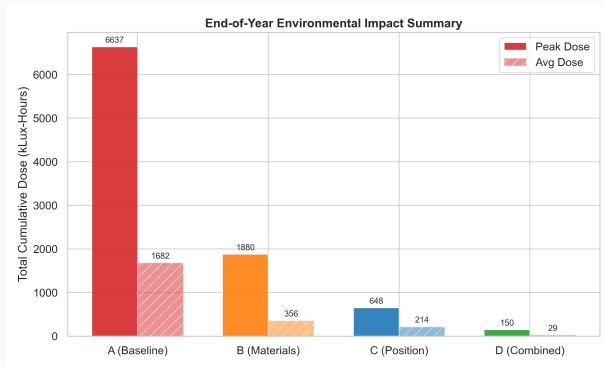
Scenario	Intervention Description
A (Baseline)	Direct southern window exposure, standard materials.
B (Materials)	UV-filtering glass and matte architectural floor applied.
C (Position)	Artifact moved to a shaded corner, preventing direct sun.
D (Synergy)	Both UV-filtering materials and corner positioning applied.

Experimental Results: Diurnal Exposure



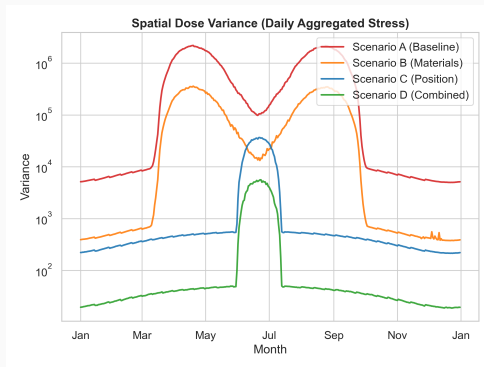
- **Solar Glare Profile:** High concentration of irradiance precisely traces the sun's path during the summer months.

Experimental Results: Final End-of-Year Dose



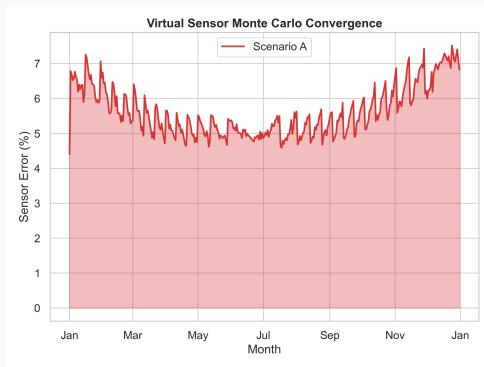
- **Massive Reduction:** Scenario D limits the absolute peak to 150k Lux-Hours (a 97.7% decrease) and average to 28k, making it safe under CIE limits.

Scientific Validation: Spatial Variance



- **Variance Constraint:** Spatial variance scales predictably with physical exposure, demonstrating a mathematically divergent-free integration across the artifact.

Scientific Validation: Sensor Error



- **Mathematical Reliability:** Stochastic Monte Carlo integration stays strictly bounded to a 4.5%–7.5% error margin against deterministic models over 8,760 hours.

- The implementation proves that Texture Space Ray Tracing is a highly viable solution for metrological dosimetry.
- Material interventions alone (Scenario B) are insufficient without geometric occlusion (Scenario C).
- Digital Twins allow safe, mathematical evaluation of conservation strategies **before** any physical intervention occurs.

- [1] M. Pharr, W. Jakob, G. Humphreys, *Physically Based Rendering*, 2016.
- [2] E. Veach, L. J. Guibas, “Optimally combining sampling techniques”, 1995.
- [3] R. Perez et al., “All-weather model for sky luminance”, 1993.